

Interpretable Neuro-Functional Music AI: A Position Paper on the *lif_lova* LIF Architecture

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Abstract

This position paper argues that the current *lif_lova* architecture represents a productive alternative to mainstream black-box music generation. Instead of treating composition as token prediction, it combines interpretable neural dynamics (leaky integrate-and-fire network states) with explicit music-theoretical constraints (functional harmony, modality, register, and style-conditioned voicing). We frame this as an *interpretable neuro-functional paradigm*: post-symbolic in state formation, symbolic in harmonic decision, and performative in real-time control. We outline why this architecture is relevant to contemporary musicology and music AI, identify current limitations, and propose a concrete research agenda with measurable evaluation criteria for interactive musical systems.

1 Thesis

The central claim is that *lif_lova* should be understood as an *instrumental AI system*, not merely a generative model. Its value lies in three properties that are often separated in current research:

1. **Causal transparency**: the pathway from signal to note is inspectable.
2. **Music-theoretical intelligibility**: harmonic behavior is articulated through explicit functional and modal logic.
3. **Performative responsiveness**: system behavior is directly controllable in live contexts.

This makes the architecture especially relevant to computational musicology, live electronic performance, and practice-based AI research.

2 Context in Music AI Research

Recent foundation-model approaches to music generation have improved stylistic plausibility and long-context prediction. However, they commonly face recurring limitations in artistic research settings:

- low interpretability of generation decisions,
- weak causal controllability during performance,
- heavy dependence on large copyrighted corpora,
- and unclear mapping between model internals and established music-theoretical constructs.

By contrast, *lif_lova* uses compact, designed dynamics rather than large corpus pretraining. The system offers explicit intermediate representations (activity bins, harmonic function assignment, modal pitch filtering, style-conditioned voicing), making analytical and pedagogical use feasible.

3 Musicological Relevance

3.1 Function as Process

In many symbolic systems, tonic/subdominant/dominant labels are imposed externally as static templates. In *lif_lova*, harmonic function is inferred from evolving neural energy distributions and local contour. This aligns with process-based musicological readings where function is enacted over time, not merely declared by chord symbols.

3.2 Modal Grammar with Dynamic Agency

The architecture constrains pitch material by per-scene modal selection (Ionian through Locrian), while timing, density, and articulation remain state-dependent. This realizes a productive split between:

- *grammar* (mode, key center, note-range framing), and
- *agency* (network-driven activation and release).

3.3 Topology as Form-Generating Constraint

Network topology (ring, fully connected, feedforward, sparse random, small-world) acts as a latent formal parameter. Each topology shapes recurrence, burst behavior, and continuity, thereby influencing perceived phrase and texture. From a musicological perspective, topology can be treated as a model of form-bearing tendency.

4 Why This Matters Beyond One System

The architecture models a broader design pattern:

continuous dynamics → interpretable musical state → performable output

This pattern may be preferable for creative AI tools where authorial control, accountability, and communicable analysis are as important as novelty.

5 Current Limitations

Despite its strengths, the present architecture has identifiable research gaps:

1. **Limited long-horizon memory:** behavior is strong at short and meso timescales but less explicit at phrase-scale planning.
2. **Crisp functional boundaries:** hard or near-hard functional transitions may reduce nuanced ambiguity.

3. **Discrete style states:** style modes are categorical rather than continuous manifolds.
4. **Evaluation gap:** current success criteria are mostly experiential and engineering-centric.

6 Research Agenda

6.1 A. Multi-timescale Formal Memory

Introduce slow control states (for example cadence pressure, repetition debt, tension trajectory) that evolve over tens of seconds or minutes. This would support longer formal arcs without abandoning local responsiveness.

6.2 B. Probabilistic Functional Inference

Replace single-label function assignment with a distribution over T/S/D classes, then smooth in time with confidence-aware hysteresis. This enables analytically meaningful functional ambiguity and gradual reinterpretation.

6.3 C. Continuous Style Manifolds

Generalize style presets into interpolable voicing spaces. Instead of switching Pop/Rock/Jazz/Blues as discrete categories, performers could navigate trajectories between them.

6.4 D. Co-Adaptive Session Learning

Add a lightweight online adaptation layer to tune density, register preference, and articulation thresholds based on performer interaction, while keeping the harmonic rule core transparent.

7 Evaluation Protocol for Interactive Music AI

We propose a mixed-method protocol that combines quantitative telemetry and qualitative judgment.

7.1 Metrics

Dimension	Example operationalization
Controllability	Parameter perturbation causes predictable output shifts
Harmonic intelligibility	Expert ratings of functional clarity and modal consistency
Temporal coherence	Segment-level continuity and phrase-shape stability
Expressive range	Coverage of density, register, and articulation states
Repeatability vs novelty	Controlled reruns with bounded divergence
Performer fit	Time-to-intent and correction burden in live sessions

Table 1: Core dimensions for evaluating instrumental AI systems.

7.2 Study Design

1. Recruit expert users (improvisers/composers) and technically proficient users.
2. Compare three conditions: fixed-rule baseline, current *lif_lova*, and enhanced variant.
3. Collect synchronized logs (control changes, topology, note events, audio descriptors).
4. Run post-session annotation interviews with timeline replay.
5. Triangulate metrics with expert panel listening.

8 Position Statement

The field should treat interpretable, performable hybrid systems as first-class music AI research objects, not as transitional artifacts between symbolic systems and large generative models. *lif_lova* demonstrates that explicit musical knowledge and neural dynamics can be co-compositional, yielding systems that are analyzable, steerable, and artistically credible.

9 Conclusion

The current *lif_lova* LIF architecture is already a compelling contribution to modern music AI discourse: it embodies an alternative research trajectory centered on transparency, theory-informed behavior, and live musical agency. The next step is not to replace this architecture with opaque end-to-end generation, but to deepen it through multi-timescale memory, probabilistic harmonic reasoning, continuous stylistic control, and rigorous interactive evaluation.

Suggested Citation Placeholder

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